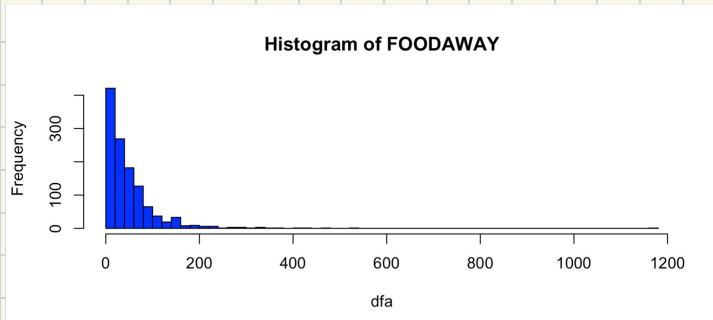


a.



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	12.04	32.55	49.27	67.50	1179.00

mean : 49.27

median : 32.55

25 percentiles : 12.04

75 percentiles : 67.50

b.

advanced degree member :

mean : 73.15

median : 48.15

no advanced or no college :

mean : 39.01

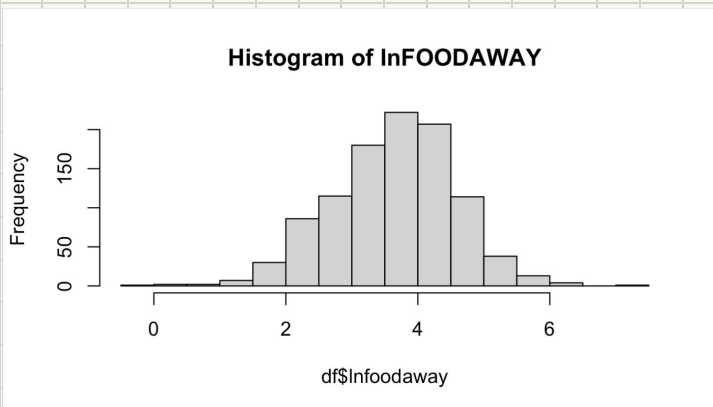
median : 26.02

college degree member :

mean : 48.60

median : 36.11

c.



number of observations :

FOODAWAY : 1200

 $\ln(\text{FOODAWAY})$: 1078

For three-person households with incomes between \$1,000 to \$20,000 per month and which FOODAWAY equals to 0, their $\ln(\text{FOODAWAY})$ diverge to negative infinity. Therefore, when plotting the histogram of $\ln(\text{FOODAWAY})$, $\text{Inf}(-\text{Inf})$ must be excluded, which result in the number of observations: $\ln(\text{FOODAWAY}) < \text{FOODAWAY}$.

d. Call:
`lm(formula = lnfoodaway ~ income, data = usable_ln)`

Residuals:

	Min	1Q	Median	3Q	Max
	-3.6547	-0.5777	0.0530	0.5937	2.7000

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.1293004	0.0565503	55.34	<2e-16 ***
income	0.0069017	0.0006546	10.54	<2e-16 ***

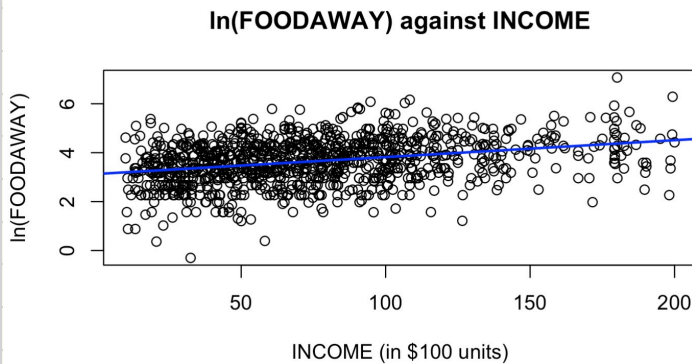
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8761 on 1020 degrees of freedom
Multiple R-squared: 0.09826, Adjusted R-squared: 0.09738
F-statistic: 111.1 on 1 and 1020 DF, p-value: < 2.2e-16

Interpret the estimated slope:

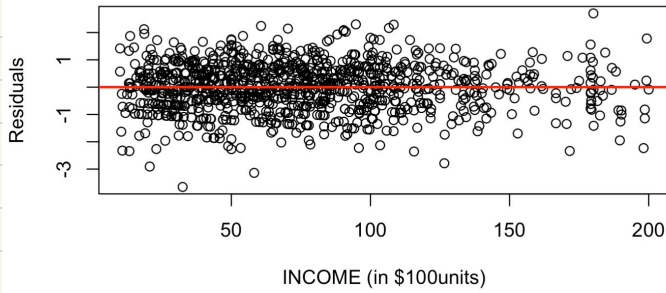
When the income increases 1 unit (= income of the household increases \$100 per month), the FOODAWAY of the household, on average, increases 0.0069017 units (= \$0.0069017).

e.



f.

Residuals vs. INCOME



The residuals are centered around zero and do not show an obvious systematic pattern, suggesting the linear specification is broadly reasonable.

Code:

```
load("/Users/liaochenghao/Desktop/Financial Econometrics/Homework/HW1/Data/cex5_small.rdata")
df <- cex5_small
names(df)

# (a)
dfa <- df$foodaway
hist(dfa, breaks = 50, main = "Histogram of FOODAWAY", col = "blue")
summary(dfa)

# (b)
df_adv <- subset(df, advanced == 1)
summary(df_adv$foodaway)
df_coll <- subset(df, college == 1)
summary(df_coll$foodaway)
df_noadv_or_nocoll <- subset(df, advanced == 0 | college == 0)
summary(df_noadvnocoll$foodaway)

# (c)
df$lnfoodaway <- log(df$foodaway)
hist(df$lnfoodaway, main = "Histogram of lnFOODAWAY")
usable_ln <- subset(df, is.finite(lnfoodaway))
sum(!is.na(df$foodaway)) # number of observation of FOODAWAY
sum(!is.na(usable_ln$foodaway)) # number of observation of ln(FOODAWAY)

# (d)
ln_reg_foodaway <- lm(lnfoodaway ~ income, data = usable_ln)
summary(ln_reg_foodaway)

# (e)
plot(usable_ln$income, usable_ln$lnfoodaway,
     xlab = "INCOME (in $100 units)", ylab = "ln(FOODAWAY)",
     main = "ln(FOODAWAY) against INCOME")
abline(ln_reg_foodaway, lwd = 2, col = "blue")

# (f)
res <- resid(ln_reg_foodaway)
plot(usable_ln$income, res,
     xlab = "INCOME (in $100units)", ylab = "Residuals",
     main = "Residuals vs. INCOME")
abline(h = 0, lwd = 2, col = "red") # Residual = 0 baseline
```